Ex 6: Programs in Prolog

1. Family Relationships

Problem: Define family relations and answer grandparent queries.

% Facts parent(john,

mary). parent(mary,

susan). parent(john,

mike).

% Rules grandparent(X, Y) :- parent(X, Z),

parent(Z, Y).

OUTPUT:

```
26 ?- consult(['c:/Users/Lenovo/Downloads/poai6.1.pl']).
true.

27 ?- grandparent(john, susan).
true
```

2: Animal Classification

Problem: Identify if an animal is a mammal based on features.

% Facts has_hair(tiger).

gives_milk(tiger).

% Rule mammal(X) :- has_hair(X),

gives_milk(X).

OUTPUT:

```
29 ?- consult(['c:/Users/Lenovo/Downloads/poai6.2.pl']).
true.

30 ?- mammal(tiger).
true.

31 ?-
```

3: Simple Arithmetic Knowledge Base

Problem: Implement simple addition using PROLOG.

% Rule add(X, Y, Z) :- Z is

X + Y.

OUTPUT:

